

DIRECTIONS

Show your work in the box for each problem.

Write your final answer on the answer line.

Use the column algorithm with regrouping or cascade as taught in lessons 3.03.01–3.03.03.

Name: _____

Date: _____

1 Addition with regrouping — line up and add

Stack the numbers by lining up the ones. Add right to left and regroup at 10.

$$7,625 + 1,835 = \underline{\hspace{2cm}}$$

SHOW YOUR STACK AND EVERY REGROUP

Sum: _____

2 Addition — two regroup

Find the sum. Show every regroup you do.

$$3,875 + 3,249 = \underline{\hspace{2cm}}$$

SHOW YOUR WORK

Sum: _____

3 Subtraction with regrouping — one unbundle

Find the difference. Show the unbundle.

$$8,420 - 3,363 = \underline{\hspace{2cm}}$$

[SHOW EVERY UNBUNDLE](#)

Difference: _____

4 Subtraction — bigger unbundle

Find the difference. Two unbundles needed.

$$5,483 - 3,397 = \underline{\hspace{2cm}}$$

[SHOW ALL YOUR WORK](#)

Difference: _____

5 Cascade through zeros

The top number has zeros. Cascade the borrow through them.

$$9,005 - 3,397 = \underline{\hspace{2cm}}$$

SHOW WHAT EACH DIGIT BECAME

Difference: _____

6 Word problem — Aki's beads

Aki had 6,302 beads. She used 1,847 beads on her sculpture.

How many beads does Aki have left?

SHOW YOUR STACK AND UNBUNDLES

Beads left: _____

7 MODEL — draw a place-value chart

Draw a place-value chart (thousands, hundreds, tens, ones) showing $7,777 + 444$.
Use it to find the sum. Mark each regroup.

DRAW YOUR CHART AND LABEL EVERY REGROUP

Sum: _____

8 EXPLAIN — what is the cascade?

Use your own words to explain why subtracting from a number with zeros (like 9,005) needs a CASCADE. What happens to each zero in the path?

WRITE 2–3 SENTENCES USING PLACE-VALUE WORDS

My explanation: _____
